

Western Mathematics

2021

YEAR 11
FINAL
EXAMINATION

Mathematics Standard

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black pen
- Approved calculators may be used
- A reference sheet is provided at the back of this paper
- In Questions in Section II, show relevant mathematical reasoning and/or calculations

Total marks : **Section I – 10 marks** (pages 2 – 5)
80

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II – 70 marks (pages 6 – 28)

- Attempt Questions 11 – 36
- Allow about 1 hour and 45 minutes for this section

Section I

10 marks

Attempt Questions 1–10

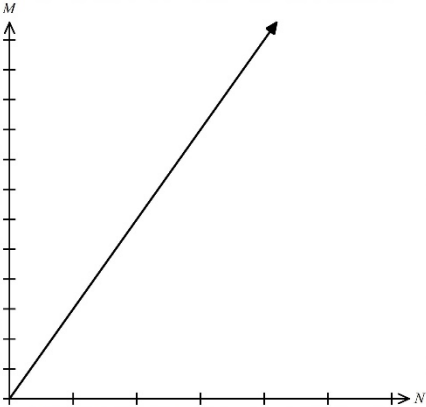
Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1 – 10

1. Jenny has a packet of 50 cards which are coloured red, yellow, blue, and green.
Thirteen of the cards are yellow and sixteen are red.
If she draws one card out at random, what is the probability that it *isn't* a yellow or red card?

- (A) $\frac{13}{50}$
(B) $\frac{16}{50}$
(C) $\frac{21}{50}$
(D) $\frac{29}{50}$

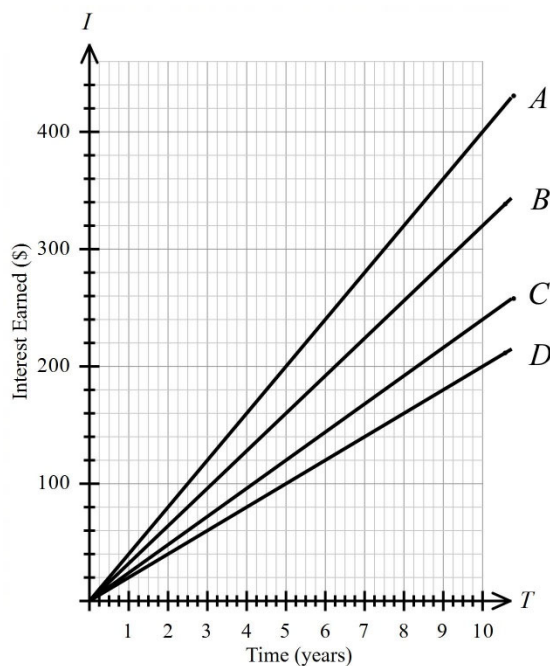
2. Two relationships between variables are represented below, one by a graph and the other by an equation.

The relationship between M and N	The relationship between P and Q
	$P = 4.5Q$

Which relationship is an example of direct variation?

- (A) Both relationships
(B) The relationship between M and N only
(C) The relationship between P and Q only
(D) Neither relationship

3. Jade measures the mass of her toddler on a digital scale which records kilograms, measured to one decimal place.
She reads the mass as 6.6 kg.
What are the limits of accuracy for this measurement?
- (A) Between 6.5 kg and 6.7 kg
(B) Between 6.55 kg and 6.65 kg
(C) Between 6.505 kg and 6.705 kg
(D) Between 6.55 kg and 6.75 kg
4. The graph below shows the amount of simple interest earned on four different investments over a ten-year period.



Which line represents an investment of \$800 at 5% p.a.?

- (A) Line A
(B) Line B
(C) Line C
(D) Line D

5. When visitors enter an Aged Care facility, as well as their name, they are asked to record three things:
- the suburb which they last visited.
 - their body temperature.
 - their age group; in these classes, Under 20, 21-40, 41-60, 61-80 and 80 and over.

Which type of data is **not** among those collected?

- (A) Categorical, nominal data
- (B) Categorical, ordinal data
- (C) Numerical, continuous data
- (D) Numerical, discrete data
6. Change the subject of the equation $m = \frac{w}{q-p}$ so that q is the subject.
- (A) $q = \frac{m+wp}{p}$
- (B) $q = m(w + mp)$
- (C) $q = \frac{m}{w+mp}$
- (D) $q = \frac{w+mp}{m}$
7. Boyd works out his family's average utility bills and plans to use this to help budget their monthly spending.

Average Household Utility Bills		
Bill	Frequency	Average amount
Electricity	Quarterly	\$535.60
Water	Quarterly	\$213.40
Phone/Internet	Monthly	\$90.60
Gas	Fortnightly	\$49.65

Based on this, how much should he put aside each month, to cover all his utilities bills for the year.

- (A) \$78.23
- (B) \$220.80
- (C) \$293.00
- (D) \$447.84

8. Abdul arrives at a bus stop at 3:19 pm.
The next three buses to his home leave at 15:25, 15:45 and 16:05 respectively.
Before he catches his bus, he has an errand that he estimates this will take half an hour.
How many minutes must he wait for a bus when he gets back to the stop?
- (A) 4 minutes
(B) 11 minutes
(C) 16 minutes
(D) 49 minutes
9. Marshall is paid an annual salary of \$124 956.00 and has 25% of his salary deducted for income tax. He also has weekly deductions of \$45.60 for union fees and \$60.50 for health insurance.
What is his weekly net pay?
- (A) \$1 696.15
(B) \$1 756.65
(C) \$1 800.72
(D) \$1 802.25
10. The table below gives the age distribution of the Australian population in 2020.

Age Group	Percentage of Australian Population
0-9	12.85%
10-19	12.46%
20-29	14.45%
30-39	14.58%
40-49	12.80%
50-59	11.96%
60-69	10.25%
70-79	7.15%
80-89	2.91%
90-99	0.56%
100+	0.02%

If eighty Australians were chosen at random, what is the expected number who would be aged less than 30?

- (A) 20
(B) 32
(C) 40
(D) 43

Western Mathematics

**2021 - YEAR 11 FINAL
EXAMINATION**

Class and Teacher

Mathematics Standard

Student Number

Section II Answer Booklet

Student Name

70 marks

Attempt Questions 11 – 36

Allow about 1 hour and 45 minutes for this section

Instructions

- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
- Your responses should include relevant mathematical reasoning and/or calculations.
- Extra writing space is provided at the back of the booklet.
If you use this space, clearly indicate which question you are answering.

Question 11 (2 marks)

2

In Australia in winter, the four eastern states (Qld, NSW, Vic and Tas) and the ACT are in the same time zone (AEST).

South Australia and the Northern Territory are half an hour behind the eastern states (ACST) and Western Australia is two hours behind them (AWST).

In summer, daylight saving is observed in New South Wales, Victoria, South Australia, Tasmania, and the ACT only.

If it is 12:15 pm in NSW on a day in January, what are the times in South Australia and Western Australia?

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Question 12 (3 marks)

(a) Solve the equation: $\frac{4p}{5} + 9 = 11$.

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(b) Given that $u = \frac{t^2+mt}{r}$, find the value of m when $u = 8$, $t = 2$ and $r = 5$.

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Question 13 (2 marks)

The table below shows the number of power poles which had to be replaced in 20 suburbs after a storm.

Number of poles (x)	No. of Suburbs Frequency (f)	Cumulative Frequency
12	2	2
13	3	5
14	2	7
15	2	9
16	3	12
17	5	17
18	3	20

(a) Find the mean number of poles. 1

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(b) What is the median number of poles? 1

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Question 14 (2 marks)

Grant bought a phone for work, which was priced at \$495 including GST, but as it was for work, he did not pay the GST.
Suzie bought the same phone, from the same store, and paid GST, but was given a 10% discount.
What is the difference in the amounts they paid for their phones?

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Question 15 (2 marks)

The table below gives the estimated number of kilojoules burned per kilogram of body weight per hour for four exercise activities.

Energy Usage in Exercise	
Activity	kJ/kg
Low Impact Aerobics	21.2
Moderate Jogging	42.4
Rowing machine	33.2
Skipping rope	52.4

- (a) Tanika who has a body weight of 48 kg, does moderate jogging for two hours.
How many kilojoules does she burn? (Answer to the nearest kJ.)

1

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Question 15 continues on page 10

Question 15 continued

- (b) Amira who has a body weight of 65 kg, skips rope for 20 minutes.
How many kilojoules does she burn? (Answer to the nearest kJ.)

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Question 16 (2 marks)

Marco bought a car for \$14 000 two years ago, and depreciated it using the straight-line method. It is now worth \$10 240, and he intends to sell it in another three years' time.

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What price should he sell it for?

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Question 17 (3 marks)

Lowitja interviewed a variety of people and released each interview as a podcast.

She recorded the cultural heritage of each guest as part of her questions.

The table below shows the number of people who identified with each category.

Group	Number
Anglo-Australian	6
Indigenous-Australian	5
Asian-Australian	4
European- Australian	5
African- Australian	2
Other	3

Mischa randomly chose one of the podcasts.

- (a) Find the probability that the interviewee is European- Australian.1

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- (b) Find the probability that the interviewee is Indigenous or African-Australian.1

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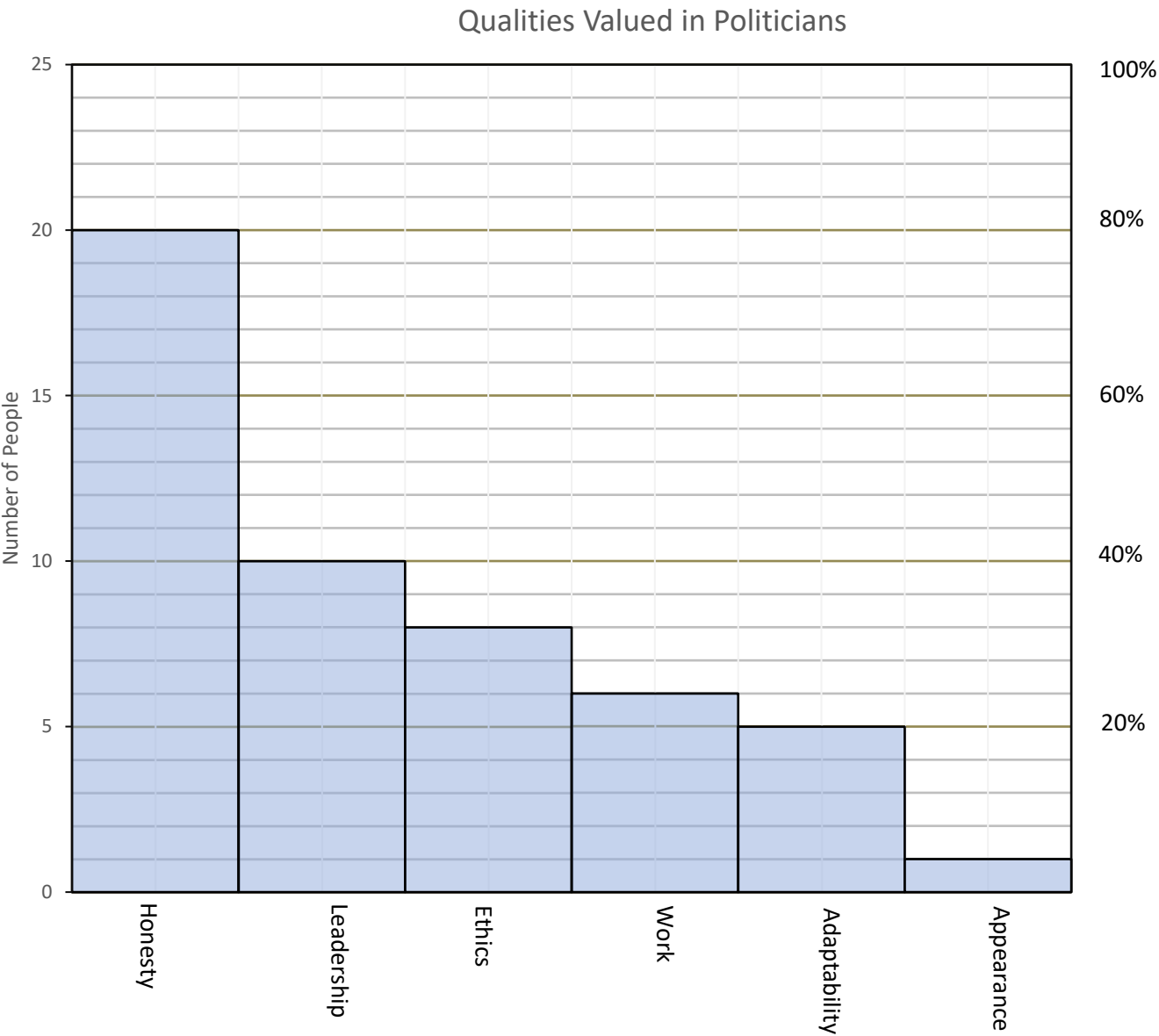
- (c) Find the probability that the interviewee is *not* Asian- Australian.1

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Question 18 (2 marks)

The histogram below shows the qualities of politicians that were valued by 50 people in a survey.



Draw a Pareto chart on the graph using the scale on the right-hand side.

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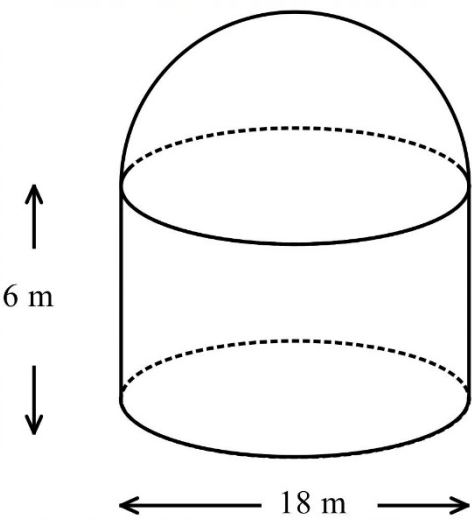
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Question 19 (3 marks)

The building for a community centre is in the shape of a cylinder, topped with a hemisphere, as shown. It has a circular concrete floor and the walls and roof are made of sheet metal.



(a) Find the area of the floor. 1

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(b) Find the external surface area of the metal walls and the roof. 2

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Question 20 (2 marks)

The table below shows the number of customers each day at a café over a month.

Class	Class Centre	Frequency
21 – 30		3
31 – 40		4
41 – 50		8
51 – 60		3
61 – 70		7
71 – 80		5

(a) Find an approximation for the mean of the data. 1

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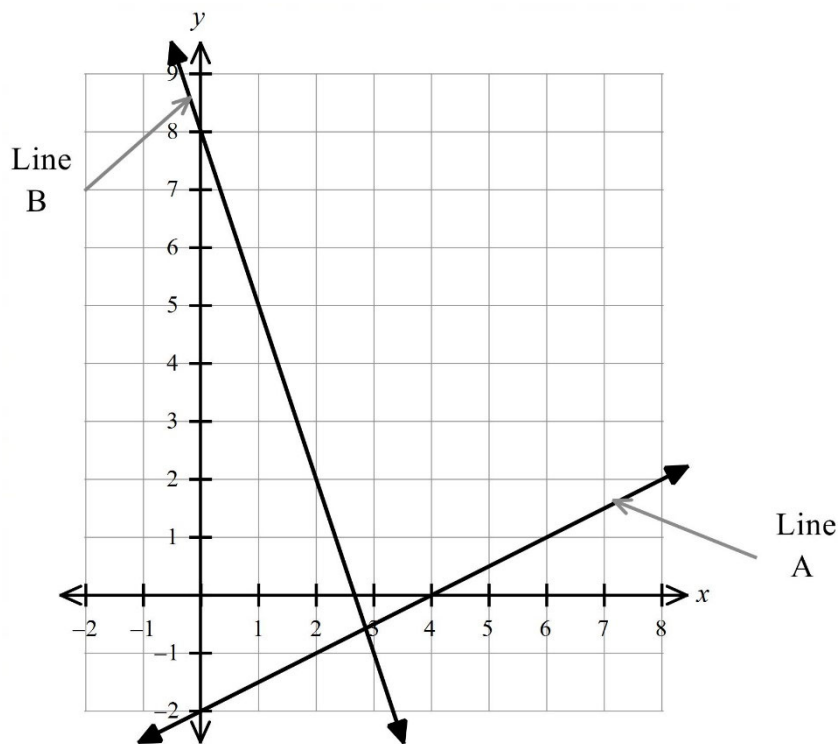
(b) Find an approximation for the standard deviation of the data. 1

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Question 21 (4 marks)

Two straight-line graphs have been drawn on the number plane below.



(a) What is the gradient of Line A? 1

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(b) Find the equation of Line B. 1

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(c) On the same graph, draw the line with equation $y = 2x + 3$ and give the point where it intersects with Line B. 2

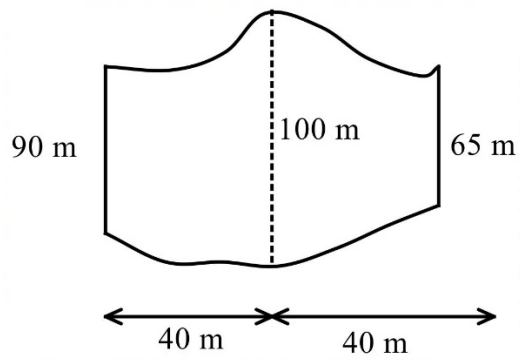
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Question 22 (2 marks)

Aiden takes the measurements shown below of an irregular field.

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Use two applications of the trapezoidal rule to estimate the area of the field.

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Question 23 (3 marks)

James is employed by a solar panel installation company and is paid a normal rate of \$45.00 per hour for an eight-hour day.

He is paid overtime at time-and-a-half for the first two hours and double time after that.

On days with extreme heat conditions, he is paid a bonus of 5% on top of his total pay.

(a) How much would he earn on a day where he works for 10 hours in extreme heat?

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(b) What is his gross pay for a week where he worked for 9 hours on two days and for 12 hours on the other three days? There was no extreme heat during the week.

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Question 24 (2 marks)

Shane owns a washing machine that is rated at 2 200 watts.

He uses it three times a week on a cycle that lasts for an hour and ten minutes.

If electricity costs 24.66 cents per kWh, what does it cost to use the washer **for a year**?

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Question 25 (2 marks)

A measurement is given as 3.4579×10^{-4} cm.

Write this as a normal decimal number, correct to 3 significant figures.

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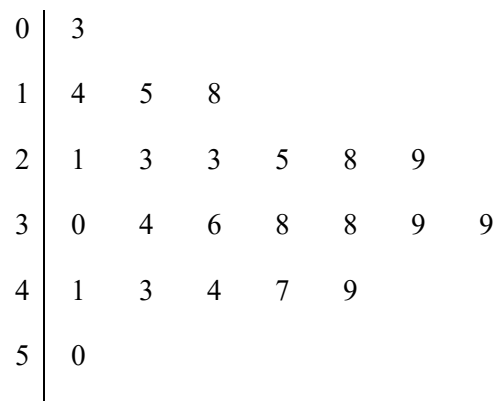
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Question 26 (3 marks)

The data in the stem-and-leaf plot below is collected as part of a research project.



(a) What is the median of the data? 1

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(b) What is the interquartile range of the data? 1

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(c) Determine if there are any outliers in the data. 1

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Question 27 (3 marks)

Susan and Adam share a bottle of wine over dinner, each consuming three standard drinks over a period of two hours.

Adam's mass is 90 kg and Susan's mass is 60 kg.

The formulae below give the BAC for males and females.

$$BAC_{\text{Male}} = \frac{10N - 7.5H}{6.8M}$$

$$BAC_{\text{Female}} = \frac{10N - 7.5H}{5.5M}$$

N – number of standard drinks

H – time in hours of consuming alcohol

M – mass in kilograms

- (a) Calculate their individual BAC readings (correct to 2 significant figures).

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- (b) Use the formula below to determine how much longer than Adam it would take for Susan's BAC to return to zero after their last drink.

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$$\text{Time(in hours)} = \frac{BAC}{0.015}$$

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Question 28 (3 marks)

Andie has an annual salary of \$98 500 and has allowable tax deductions of \$460, \$780, and \$3 500.

(a) Use the table below to calculate her income tax due for the year.

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Taxable Income	Tax On This Income
\$0 to \$18,200	Nil
\$18,201 to \$45,000	19c for each \$1 over \$18,200
\$45,001 to \$120,000	\$5,092 plus 32.5c for each \$1 over \$45,000
\$120,001 to \$180,000	\$29,467 plus 37c for each \$1 over \$120,000
\$180,001 and over	\$51,667 plus 45c for each \$1 over \$180,000

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(b) Andie calculates her annual Medicare levy to be \$1 875.20.

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She is paid fortnightly and has \$880.50 taken out in PAYG instalments.

How much should she get back as a tax refund at the end of the tax year?

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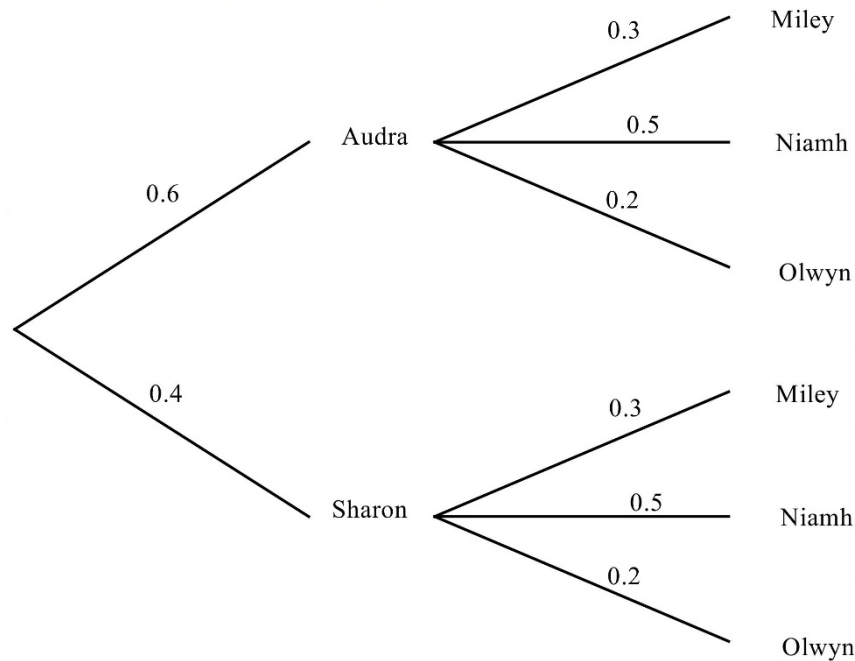
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Question 29 (4 marks)

In a tennis tournament there are five players on a team and two games of singles are to be played.

Either Audra or Sharon will play in the senior singles and one of Miley, Niamh or Olwyn will play in the junior singles.

The probability tree diagram below shows the likelihood of each player being chosen, by the coach.



(a) What is the probability that Audra and Niamh will both play singles? 1

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(b) What is the probability that Sharon or Olwyn or both will play singles? 2

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(c) What is the probability that neither Audra nor Miley will play? 1

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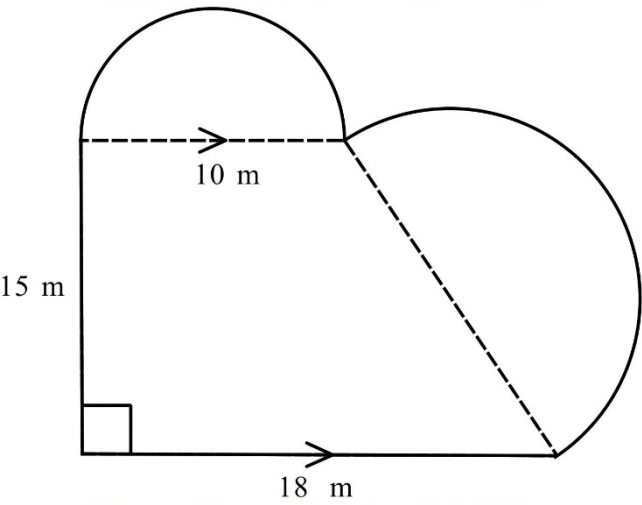
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Question 30 (2 marks)

Find the perimeter of the figure below, correct to the nearest 10th of a metre.

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Question 31 (3 marks)

Zoe is a nurse, and she regularly uses Young's formula to calculate drug dosages for children aged 1 – 12 years.

$$\text{Childs Dosage} = \frac{\text{age of child (in years)} \times \text{adult dosage}}{\text{age of child (in years)} + 12}.$$

- (a) Harry is 6 years old and is prescribed Corozone, which has an adult dosage of 7.5 ml of liquid formulation. 1

What dosage should he receive?

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- (b) Katrina is 8 years old and is prescribed Serafine with an adult dose of two 28 mg tablets. Serafine is also available in 8 mg and 17 mg tablets. All three dosages of tablet can be cut in half. 2

What combination of tablets would give her correct dose, to the nearest mg?

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Question 32 (3 marks)

Peta is buying a new car which has a purchase price of \$55 730 plus on-road costs.

The on-road costs are as follows:

- Registration: \$675.50.
- Stamp duty: \$1350 plus \$5 per \$100 or part thereof on selling price.
- Green slip insurance: \$342.50.
- Comprehensive car insurance: \$1 250, less a 20% loyalty discount.

(a) Calculate the Stamp duty on the vehicle. 1

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(b) Find the cost of the comprehensive car insurance. 1

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(c) What will it cost her in total to own the car and drive it away, fully insured? 1

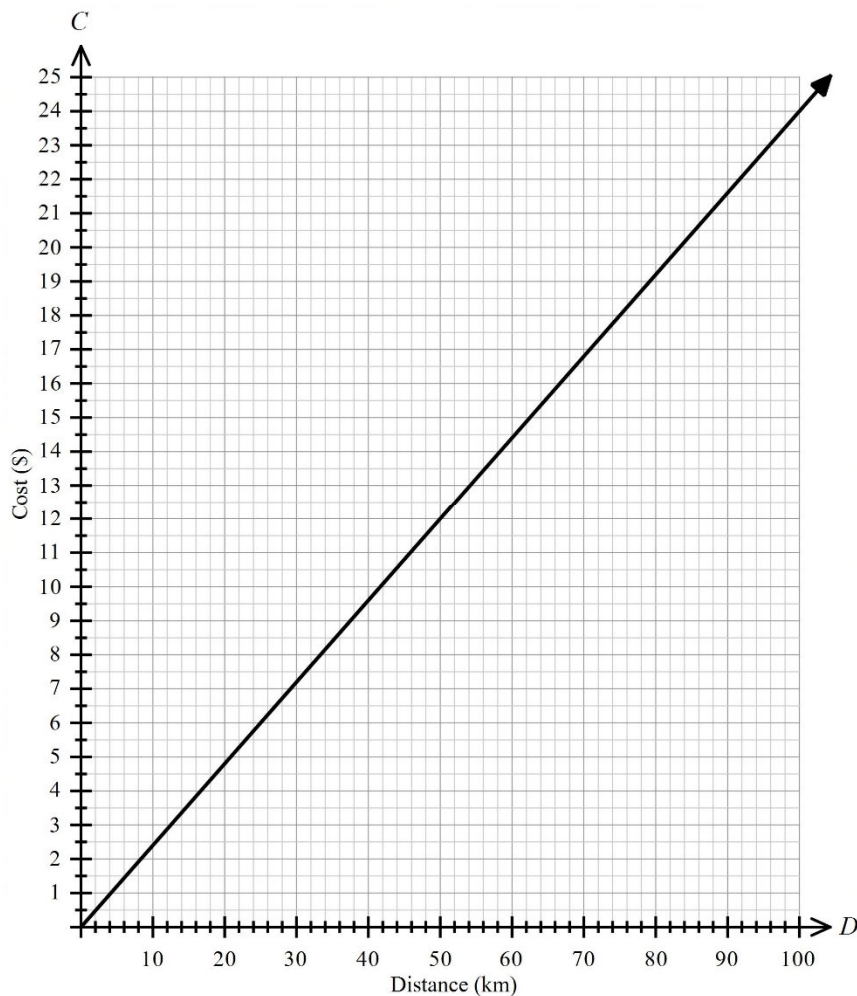
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Question 33 (3 marks)

Lucy draws the graph below to show the running cost per km of her car.



- (a) What would it cost for a journey of 75 km?

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- (b) Ayla calculates the running cost of her vehicle is \$0.20 per km.

1

Draw a line on the grid above to represent the running cost of Ayla's car.

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- (c) How much more would it cost to travel in Lucy's car for a 50 km trip compared to Ayla's?

1

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Question 34 (3 marks)

Tahlia has a job with a nett pay of \$880.00 per week which is paid into her bank account.

To budget her spending, Tahlia divides her bank account into the five sub-accounts below and she will allocate a portion of her weekly pay to each of these.

Personal Spending	Energy and Accommodation	Savings
Entertainment and holidays	Car costs	

- (a) The current rent on her unit is \$900.00 per fortnight and the energy bills for the unit total \$485.00 each quarter, on average. 1

She shares both these expenses equally with her flatmate.

How much should she allocate to the Energy and Accommodation sub-account each week?

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- (b) Her average monthly personal spending in the last twelve months was \$1 859.00. 1
- She wants to maintain this level in the next year.

How much should she allocate to the Personal Spending sub-account each week?

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- (c) She also calculated the amount to allocate to the following sub-accounts each week. 1

Car costs - \$68.00

Entertainment and Holidays - \$108.00

If she allocates these amounts and those in parts (a) and (b), how much will be left for the saving sub-account each week?

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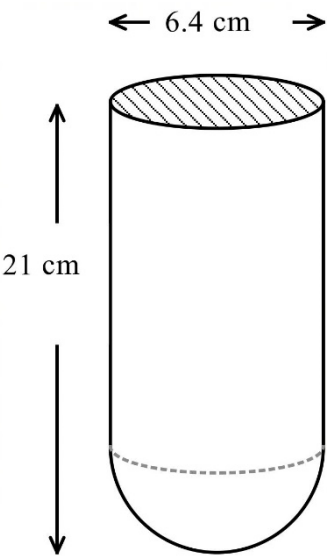
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Question 35 (3 marks)

A receptacle for windscreen washer fluid in a vehicle is in the shape of a cylinder with a hemisphere at one end as shown.



- (a) Calculate the volume of the receptacle in cubic centimetres. 2

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- (b) An amount of 0.02 litres of fluid is drawn out from the bottom of the receptacle each time a lever is pressed. Every cubic centimetre of volume holds one millilitre. 1

How many times could the lever be pressed before the receptacle is empty?

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Question 36 (4 marks)

Jennifer is living in Dubbo (UTC+10) and her two friends, Samantha and Marissa are volunteering overseas.

Samantha is in Brasilia (UTC-3) and Marissa is in Khartoum (UTC +2).

- (a) What is the time difference between Jennifer and Samantha?

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- (b) At 4 pm local time in Dubbo, Jennifer rings Marissa.

1

What is the local time that the call is received?

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- (c) Samantha flies out of Brasilia on a ten-hour flight to Khartoum.

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She leaves at 12 noon on 15th August and when she arrives, she calls Jennifer immediately.

What is the local time that Jennifer receives the call?

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End of Examination

Section II Extra writing space

If you use this space, clearly indicate which question you are answering.

[illegible]

Mathematics Standard 1
Mathematics Standard 2

REFERENCE SHEET

Measurement

Limits of accuracy

$$\text{Absolute error} = \frac{1}{2} \times \text{precision}$$

$$\text{Upper bound} = \text{measurement} + \text{absolute error}$$

$$\text{Lower bound} = \text{measurement} - \text{absolute error}$$

Length

$$l = \frac{\theta}{360} \times 2\pi r$$

Area

$$A = \frac{\theta}{360} \times \pi r^2$$

$$A = \frac{h}{2}(a + b)$$

$$A \approx \frac{h}{2}(d_f + d_l)$$

Surface area

$$A = 2\pi r^2 + 2\pi rh$$

$$A = 4\pi r^2$$

Volume

$$V = \frac{1}{3}Ah$$

$$V = \frac{4}{3}\pi r^3$$

Trigonometry

$$\sin A = \frac{\text{opp}}{\text{hyp}}, \quad \cos A = \frac{\text{adj}}{\text{hyp}}, \quad \tan A = \frac{\text{opp}}{\text{adj}}$$

$$A = \frac{1}{2}ab \sin C$$

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

Financial Mathematics

$$FV = PV(1 + r)^n$$

Straight-line method of depreciation

$$S = V_0 - Dn$$

Declining-balance method of depreciation

$$S = V_0(1 - r)^n$$

Statistical Analysis

An outlier is a score

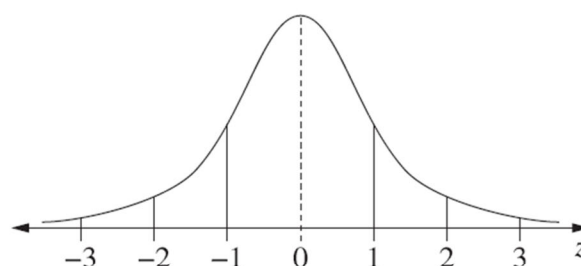
$$\text{less than } Q_1 - 1.5 \times IQR$$

or

$$\text{more than } Q_3 + 1.5 \times IQR$$

$$z = \frac{x - \mu}{\sigma}$$

Normal distribution



- approximately 68% of scores have z -scores between -1 and 1
- approximately 95% of scores have z -scores between -2 and 2
- approximately 99.7% of scores have z -scores between -3 and 3

Western Mathematics
2021 Year 11 Final Examination
Mathematics Standard

Name _____ Teacher _____

Section I – Multiple Choice Answer Sheet

Allow about 15 minutes for this section.

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
 A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word **correct** and drawing an arrow as follows.

A ☒ B ☒ ^{correct} C ☐ D ☐

- | | | | | | | | | |
|-----|---|-----------------------|---|-----------------------|---|-----------------------|---|-----------------------|
| 1. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 2. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 3. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 4. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 5. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 6. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 7. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 8. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 9. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 10. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |